

KENDRIYA VIDYALAYA SANGATHAN, AGRA REGION

FIRST PRE BOARD EXAMINATION 2024-25

CLASS : XII

SUBJECT: PHYSICS (THEORY)

Maximum Marks: 70

Time Allowed: 3 hours.

General Instructions:

- (1) There are 33 questions in all. All questions are compulsory.
- (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- (3) All the sections are compulsory.
- (4) Section A contains sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, Section B contains five questions of two marks each, Section C contains seven questions of three marks each, Section D contains two case study based questions of four marks each and Section E contains three long answer questions of five marks each.
- (5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
- (6) Use of calculator is not allowed.
- (7) You may use the following values of physical constants where ever necessary

i. $c = 3 \times 10^8 \text{ m/s}$

ii. $m_e = 9.1 \times 10^{-31} \text{ kg}$

iii. $m_p = 1.7 \times 10^{-27} \text{ kg}$

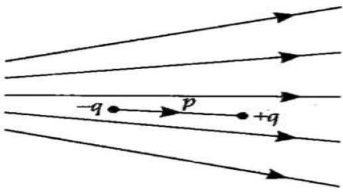
iv. $e = 1.6 \times 10^{-19} \text{ C}$

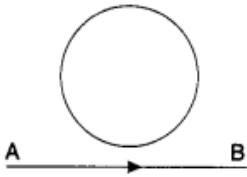
v. $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$

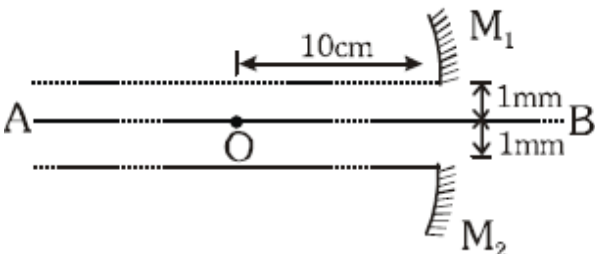
vi. $h = 6.63 \times 10^{-34} \text{ Js}$

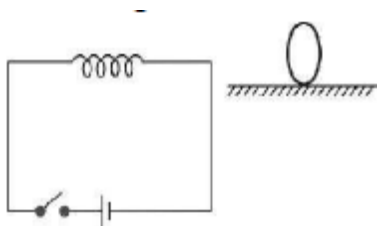
vii. $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

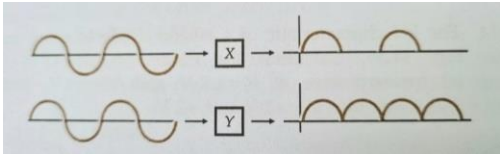
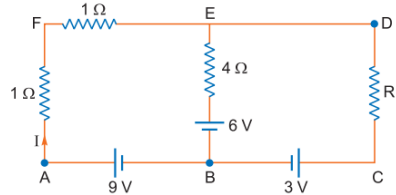
viii. Avogadro's number = 6.023×10^{23} per gram mole

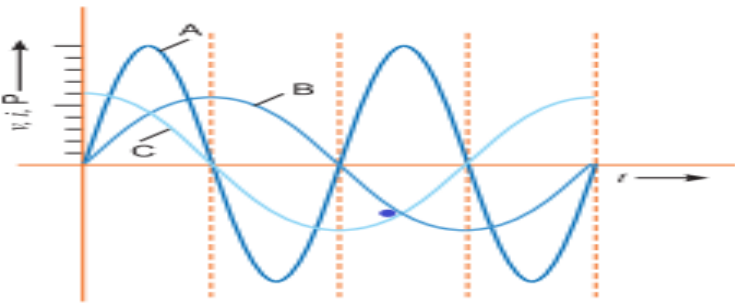
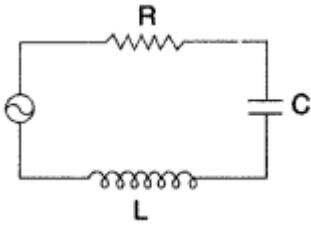
	SECTION A	
1	Figure here shows electric field lines in which an electric dipole $\mathbf{p}$ is placed as shown. Which of the following statements is correct? (a) The dipole will not experience any force. (b) The dipole will experience a force towards right. (c) The dipole will experience a force towards left. (d) The dipole will experience a force upwards. 	1
2	A small metallic ball having charge q is placed inside an insulated box. The Insulated box is then placed inside a metallic box. The net charge enclosed inside the metallic box is: (a) Zero (b) q (c) $-q$ (d) $2q$	1
3	Two wires A and B of the same material, having radii in the ratio 1:2 and carrying currents in the ratio 4:1. The ratio of drift velocity of electrons in A & B is (a) 16:1 (b) 1:16 (c) 1:4 (d) 4:1	1
4	Two long conductors separated by a distance d carry currents I_1 and I_2 in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to $3d$. The new value of the force between them is	1

	(a) $-2F$ (b) $F/3$ (c) $-2F/3$ (d) $-F/3$	
5	A short bar magnet free to rotate on horizontal plane has a magnetic moment of 0.48 J/T . Magnetic field produced by the magnet at a distance of 10 cm from the centre of the magnet on the equatorial lines (normal bisector) of the magnet has a direction and magnitude of (a) $0.43 \times 10^{-4} \text{ T}$ along N-S direction (b) $0.38 \times 10^{-4} \text{ T}$ along N-S direction (c) $0.55 \times 10^{-4} \text{ T}$ along N-S direction (d) $0.48 \times 10^{-4} \text{ T}$ along N-S direction	1
6	In the given figure current from A to B in the straight wire is decreasing. The direction of induced current in the loop is A (a) clockwise (b) anticlockwise (c) changing (d) nothing can be said 	1
7	The correct option, if speeds of gamma rays, X-rays and microwave are V_g , V_x and V_m respectively in space will be. (a) $V_g > V_x > V_m$ (b) $V_g < V_x < V_m$ (c) $V_g > V_x > V_m$ (d) $V_g = V_x = V_m$	1
8	For the same value of angle of incidence, the angles of refraction in three media A, B and C are 15° , 25° and 35° respectively. In which medium would the velocity of light be minimum? (a) A (b) B (c) C (d) Same in all media.	1

9	If the 8th bright fringe of light of wavelength λ_1 coincides with the 9th bright fringe of light of the wavelength λ_2 in young's double experiment, then the possible wavelengths of visible light are (a).400nm and 450nm (b).425nm and 400nm (c).400nm and 425 nm (d). 450nm and 400nm	1
10	Light of frequency 1.9 times the threshold frequency is incident on a photosensitive material. If the frequency is halved and intensity is doubled, the photocurrent becomes (a) doubled (b) quadrupled (c) halved (d) zero	1
11	A nucleus of mass number 189 splits into two nuclei having mass no125 and 64.the ratio of radius of daughter nuclei respective is (a) 25:16 (b) 1:1 (c) 4:5 (d) 5:4	1
12	A concave mirror of focal length 20 cm is cut into two parts from the middle and the two parts are moved perpendicularly by a distance 1 mm from the previous principal axis AB. The distance between the images formed by the two pars is  (a) 2 mm (b) 6 mm (c) 3 mm (d) 4 mm	1

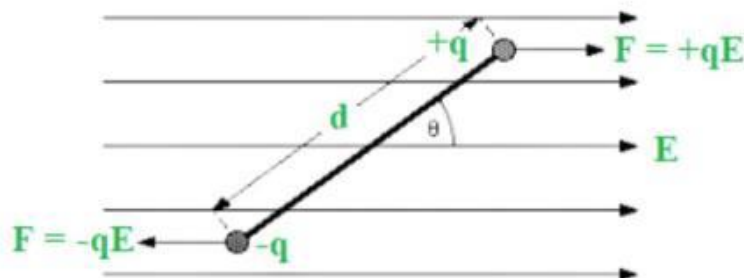
	ASSERTION REASON QUESTIONS Two statements are given – one labeled Assertion (A) and other labeled as Reason (R). Select the correct answer to these questions from the codes (A),(B),(C),(D) as given below. (A). Both A and R are true and R is the correct explanation of A (B). Both A and R are true and R is not the correct explanation of A (C). A is true but R is false (D). A is false and R is also false	
13	Assertion(A) : Figure shows a horizontal solenoid connected to a battery and a switch. A copper ring is placed on a smooth surface, the axis of the ring being horizontal. As the switch is closed, the ring will move away from the solenoid.  Reason(R) : Induced emf in the ring, $e = - d\Phi/dt$	1
14	Assertion (A): The maximum intensity in interference pattern is four times the intensity due to each slit. Reason (R): Intensity is directly proportional to square of amplitude	1
15	Assertion(A): The diffusion current in a p-n junction is from the p-side to the n-side. Reason(R): The diffusion current in a p-n junction is greater than the drift current when the junction is in forward biased.	1
16	Assertion(A): Increasing the charge on the plates of a capacitor means increasing the capacitance. Reason(R): Capacitance is directly proportional to charge.	1

	SECTION B	
17	An a.c signal is fed into two circuits X and Y and the corresponding output in the two cases have the wave fronts shown in fig given below .Name the circuit X and Y .Also draw either X or Y detailed circuit diagram. 	2
18	The ground state energy of hydrogen atom is -13.6 eV. What are the kinetic and potential energies of the electron in this state?	2
19	Calculate the speed of light in a medium whose critical angle is 45°. Does critical angle for a given pair of media depend on the wavelength of incident light? Give reason. Or Find the intensity at a point on the screen in Young's double slit experiment where the interfering waves of equal intensity I each have a path difference of $\lambda/3$	2
20	A galvanometer of resistance 120 ohm gives full scale deflection for a current of 5mA. How can it be converted into an ammeter of range 0 to 5A? Also determine the net resistance of the ammeter.	2
21	A cell of emf E and internal resistance r is connected across an external resistor R . write the relation between E and r . Plot the graph showing the variation of Potential difference V across R , versus resistance R .	2
	SECTION C	
22	Using Kirchhoff's rules determine the value of unknown resistance R in the circuit so that no current flows through $4\ \Omega$ resistances. Also find the potential difference between A and D. 	3

23	A parallel plate capacitor of capacitance C is charged to a potential V by a battery. Without disconnecting the battery, the distance between the plates is tripled and dielectric medium of $K = 10$ is introduced between plates of capacitor. Explain giving reason how will the following be affected? (i) Capacitance of capacitor (ii) Charge on capacitor (iii) Energy density of capacitor	3
24	A device 'X' is connected to an ac source. The variation of voltage, current and power in one complete cycle is shown in the figure  (i) Identify the device X. (ii) Which of the curves A, B and C represent the voltage, current and the power consumed in the circuit? Justify the answer. (iii) How does its impedance vary with frequency of the AC source? Show graphically. OR (a) The figure shows a series LCR circuit with $L = 5.0 \text{ H}$, $C = 80 \mu\text{F}$, $R = 40 \Omega$ connected to a variable frequency 240 V source. Calculate (i) The angular frequency of the source which drives the circuit at resonance. (ii) The current at the resonating frequency. (iii) The rms potential drop across the capacitor at resonance. 	3

25	Name the following parts of the electromagnetic spectrums. - Used in radar systems for aircraft navigation. - Used to treat muscular strain. - Used in hospitals for diagnosing diseases. Briefly describe how these waves can be produced.	3
26	(a) A capacitor is connected in series to an ammeter across a d.c. source. Why does the ammeter show a momentary deflection during the charging of the capacitor? What would be the deflection when it is fully charged? (b) How is the generalized form of Ampere's circuital law obtained to include the term due to displacement current?	3
27	Yellow light ($\lambda = 6000\text{\AA}$) illuminates a single slit of width $1 \times 10^{-4} \text{ m}$. Calculate - the distance between the two dark lines on either side of the central maximum, when the diffraction pattern is viewed on a screen kept 1.5 m away from the slit; - the angular spread of the first diffraction minimum.	3
28	Using the graph shown in the figure for stopping potential v/s the incident frequency of photons, calculate Planck's constant.	3
	SECTION D (CASE BASED QUESTION)	
29	When electric dipole is placed in uniform electric field, its two charges experience equal and opposite forces, which cancel each other and hence net force on electric dipole in uniform electric field is zero. However these	4

forces are not collinear, so they give rise to some torque on the dipole. Since net force on electric dipole in uniform electric field is zero, so no work is done in moving the electric dipole in uniform electric field. However, some work is done in rotating the dipole against the torque acting on it.



Electric Dipole in External Field

1. The dipole moment of a dipole in a uniform external field $\mathbf{E}$ is $\mathbf{P}$. Then the torque $\boldsymbol{\tau}$ acting on the dipole is

(a) $\boldsymbol{\tau} = \mathbf{P} \times \mathbf{E}$ (b) $\boldsymbol{\tau} = \mathbf{P} \cdot \mathbf{E}$ (c) $\boldsymbol{\tau} = 2(\mathbf{P} + \mathbf{E})$ (d) $\boldsymbol{\tau} = (\mathbf{P} + \mathbf{E})$

2. An electric dipole consists of two opposite charges, each of magnitude $1.0\mu\text{C}$ separated by a distance of 2.0 cm . The dipole is placed in an external field of 10^5 NC^{-1} . The maximum torque on the dipole is

(a) $0.2 \times 10^{-3}\text{ Nm}$ (b) $1 \times 10^{-3}\text{ Nm}$ (c) $2 \times 10^{-3}\text{ Nm}$ (d) $4 \times 10^{-3}\text{ Nm}$

3. Torque on a dipole in a uniform electric field is minimum when θ is equal to

(a) 0° (b) 90° (c) 180° (d) Both (a) and (c)

4. When an electric dipole is held at an angle in a uniform electric field, the net force F and torque τ on the dipole are

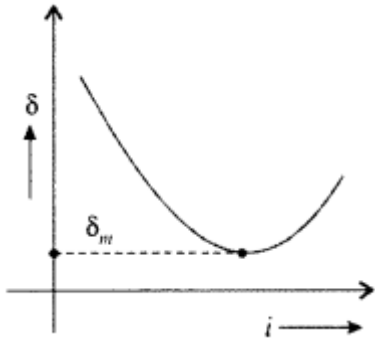
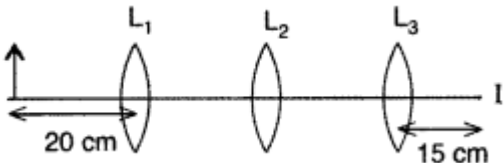
(a) $F = 0, \tau = 0$ (b) $F \neq 0, \tau \neq 0$ (c) $F = 0, \tau \neq 0$ (d) $F \neq 0, \tau = 0$

OR

4. An electric dipole of moment p is placed in an electric field of intensity E . The dipole acquires a position such that the axis of the dipole makes an angle with the direction of the field. Assuming that potential energy of the dipole to be zero when $\theta = 90^\circ$, the torque and the potential energy of the

	dipole will respectively be (a) $pE\sin\theta$, $-pE\cos\theta$ (b) $pE\sin\theta$, $-2pE\cos\theta$ (c) $pE\sin\theta$, $2pE\cos\theta$ (d) $pE\cos\theta$, $-pE\sin\theta$	
30	Motions of the charge carriers – if you burst a helium -filled balloon, helium atoms will diffuse (spread) outward into the surrounding air. This happens because there are very few helium atoms in normal air. There is a helium density gradient at the balloon- air interface (the number density of helium atoms varies across the interface), the helium atoms move so as to reduce the gradient. In same way, electrons on the n-side are close to the junction plane tend to diffuse across it and into the p-side, where there are very few free electrons. Similarly, holes on the p-side are close to the junction plane tend to diffuse across that plane and into the n-side, where there are very few holes. The motions of both the electrons and the holes contribute to diffusion current ($I_{diff.}$). The diagram illustrates a P-N Junction Diode. On the left is the P-type region, labeled 'P-type Holes', containing small circles representing holes. On the right is the N-type region, labeled 'N-type Electrons', containing small dots representing electrons. In the center, a vertical line represents the junction. Above this line, an arrow points to the 'Depletion region'. Within this region, the P-side is marked with '+' signs and the N-side with '-' signs, indicating the net positive and negative charges respectively after the initial diffusion of carriers. 1. Silicon is doped with which of the following to obtain p-type semiconductor? - Phosphorus - Gallium - Germanium - Bismuth 2. A semiconductor has an electron concentration of 6×10^{22} per m^3 and hole concentration of 8.5×10^9 per m^3. Then it is	4

	a. n-type conductor b. p-type conductor c. intrinsic conductor d. conductor 3. In a p-n-junction diode a. The current in the reverse biased condition is generally very small $\sim \mu\text{A}$. b. The current in the reverse biased condition is small but the forward biased current is independent of the bias voltage. c. The reverse biased current is strongly dependent on the applied voltage. d. The forward biased current is very small in comparison to reverse biased current. 4. In the middle of the depletion layer of a reverse biased p-n junction, the a. Electric field is zero b. Potential is maximum c. Electric field is maximum d. Potential is zero OR The dominant mechanism for the motion of charge carriers in forward and reverse biased silicon junctions are a. Drift in forward bias, diffusion in reverse bias b. Diffusion in forward bias, drift in reverse bias c. Diffusion in both forward and reverse bias d. Drift in both forward and reverse bias.	
	SECTION E	
31	(a) Sketch the magnetic field for a circular current loop and indicate the direction of the field. (b) State using a suitable diagram, the working principle of a moving coil galvanometer. What is the function of a radial magnetic field and the soft	5

	iron core used in it? (c) For converting a galvanometer into an ammeter, a shunt resistance of small value is used in parallel, whereas in the case of a voltmeter a resistance of large value is used in series. Explain why OR (a) State Biot-Savart law. Deduce the expression for the magnetic field due to a circular current carrying loop at a point lying on its axis. (b) Derive the expression for the torque on a rectangular current carrying loop suspended in a uniform magnetic field. (c) A proton and a deuteron having equal momentum enter in a region of uniform magnetic field at right angle to the direction of the field. Depict their trajectories in the field.	
32	(a) The relation, between the angle of incidence (i) and the corresponding, angle of deviation (δ), for a certain optical device, is represented by the graph shown in the figure. Identify this device. Draw a ray diagram for this device and use it for obtaining an expression for the refractive index of the material of this device in terms of an angle characteristic of the device and the angle, marked an δ_m, in the graph.  (b) You are given three lenses L_1, L_2 and L_3 each of focal length 15 cm. An object is kept at 20 cm in front of L_1, as shown. The final real image is formed at the focus 'I' of L_3. Find the separations between L_1, L_2 and L_3. 	5

	OR (a) Define power of a lens. Write its S.I units. Two thin convex lenses of focal lengths f_1 and f_2 are placed in contact coaxially. Derive the expression for the effective focal length of the combination. (b) A small telescope has an objective lens of focal length 140 cm and an eye-piece of focal length 5.0 cm. Find the magnifying power of the telescope for viewing distant objects when (i) the telescope is in normal adjustment, (ii) the final image is formed at the least distance of distinct vision.	
33	(a) Draw a schematic arrangement of Geiger- Marsden experiment showing the scattering of α-particles by a thin foil of gold. Why is it that most of the α -particles go right through the foil and only a small fraction gets scattered at large angles? Draw the trajectory of the α -particle in the coulomb field of a nucleus. What is the significance of impact parameter and what information can be obtained regarding the size of the nucleus? (b) Estimate the distance of closest approach to the nucleus ($Z = 80$) if a 7.7 MeV α-particle before it comes momentarily to rest and reverses its direction OR (a) Draw the plot of binding energy per nucleon (BE/A) as a function of mass number A. Write two important conclusions that can be drawn regarding the nature of nuclear force. (b) Use this graph to explain the release of energy in both the processes of nuclear fusion and fission (c) Name the strongest force in nature .write two characteristic of this force .	5

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